Analysys of Quantization Error

April 21, 2026

Contents

1 Error Analysis for Quantization in Linear Layers **2**
- 1.1 Linear Layers . . . 2
- 1.2 Models . . . 2
- 1.3 Error sources . . . 2
  - 1.3.1 Parameter Quantization . . . 2
  - 1.3.2 Input Quantization . . . 3
  - 1.3.3 Rescaling . . . 4
  - 1.3.4 Clipping . . . 4
- 1.4 Computing error for a linear layer . . . 4
- 1.5 Total error propagation . . . 5

2 Augmenting models with non-linear layers **5**
- 2.1 ReLU . . . 5
- 2.2 Others . . . 6

3 Annexes **6**
- 3.1 Proof of error bounds on a single layer . . . 6
  - 3.1.1 Rescaling error . . . 6
  - 3.1.2 Accumulated error $\Delta z^{(n)}$ . . . 6
  - 3.1.3 Final result . . . 7

1 Error Analysis for Quantization in Linear Layers

1.1 Linear Layers

We simplify linear layers to operation:

$$z = W \cdot x$$

To add the bias in the error analysis isn't hard.

1.2 Models

- Description of how the model behaves through an inference
- We simulate models containing a sequence of L linear layers
- $x^{(0)}$ the original input and $z^{(L)}$ final output
- Quantization bring new steps
  - Quantization of the input
  - After matrix mult, the output is rescaled to fit the chosen quantization range

Base model

$$x^{(0)} \xrightarrow{\text{linear}} z^{(0)} \xrightarrow{=} x^{(1)} \to z^{(1)} \to \ldots \to z^{(L)}$$

Quantized model

$$x^{(0)} \xrightarrow{\text{quantization}} x_{int}^{(0)} \xrightarrow{\text{linear}} z_{int}^{(0)} \xrightarrow{\text{rescaling}} x_{int}^{(1)} \to z_{int}^{(1)} \to \ldots \to z_{int}^{(L)} \xrightarrow{\text{dequantization}} \hat{z}^{(L)}$$

Virtual quantized model converted back to floats for comparison with base model

$$x^{(0)} \xrightarrow{\text{quantization}} \hat{x}^{(0)} \xrightarrow{\text{linear}} \hat{z}^{(0)} \xrightarrow{\text{rescaling}} \hat{x}^{(1)} \to \hat{z}^{(1)} \to \ldots \to \hat{z}^{(L)}$$

1.3 Error sources

Quantization introduce approximation error when:

- quantizing input,
- quantizing parameters
- rescaling output of any matrix multiplication to keep them within quantization range
- clipping when value overflows the designated range
- non-linear layers approx (see in next section)

1.3.1 Parameter Quantization

The basic quantization relationship for parameters W:

Definition

- W is the parameter of base model
- W_{int} is the quantized parameter
- $\hat{W}$ is the quantized parameter converted back to floats for comparison with base model W
- S_W is the scaling factor for W and is computed once during the quantization process

$$W_{\text{int}} = round\left(\frac{W}{S_W}\right)$$

$$\hat{W} = W_{\text{int}} \cdot S_W = round\left(\frac{W}{S_W}\right) \cdot S_W$$

Bounds Special property between base and quantized parameters.

$$\forall i, j \quad \Delta W_{ij} = \left\| W_{ij} - \hat{W}_{ij} \right\|_{\infty} \leq \frac{S_W}{2} \tag{1}$$

Proof in Annex. (TODO)

1.3.2 Input Quantization

Definition

- this is a special case of rescaling
- x is the input of base model
- x_{int} is the quantized input
- $\hat{x}$ is the quantized input converted back to floats for comparison with base model x
- $S_x^{(0)}$ is the scaling factor for x and is *computed once during a calibration phase*

$$x_{int} = round\left(\frac{x}{S_x^{(0)}}\right)$$

$$\hat{x} = x_{int} \cdot S_x = round\left(\frac{x}{S_x^{(0)}}\right) \cdot S_x^{(0)}$$

Bounds

$$\Delta x = \|x - \hat{x}\|_{\infty} \leq \frac{S_x^{(0)}}{2} \tag{2}$$

Proof This is a property of the rounding operation.

$$|a - round(\frac{a}{b}) \cdot b| \leq \frac{b}{2}$$

1.3.3 Rescaling

- Happens after linear layers $z^{(n)} = W^{(n)} \cdot x^{(n)}$
- output $z^{(n)}$ is at scale $S_W^{(n)} \cdot S_x^{(n)}$ and must scaling must be reset
- then scale back to designated range with $S_x^{(n+1)}$
- $S_x^{(n+1)}$ is the scaling factor for this linear layer and is *computed once during a calibration phase*

$$x_{int}^{(n+1)} = round\left(\frac{z_{int}^{(n)} \cdot S_W^{(n)} \cdot S_x^{(n)}}{S_x^{(n+1)}}\right) = round\left(\frac{\hat{z}^{(n)}}{S_x^{(n+1)}}\right)$$

$$\hat{x}^{(n+1)} = x_{int}^{(n+1)} \cdot S_x^{(n+1)} = round\left(\frac{\hat{z}^{(n)}}{S_x^{(n+1)}}\right) \cdot S_x^{(n+1)}$$

Bounds

$$\Delta x^{(n)} = \left\| x^{(n)} - \hat{x}^{(n)} \right\|_\infty \leq \frac{S_x^{(n)}}{2} \tag{3}$$

Note

- Input quantization is a special case of rescaling
- that's why we name rescale factor for input as $S_x^{(0)}$

1.3.4 Clipping

- Happens if some value overflows from designated range
- this should not happen if scale factors are chosen correctly
- but on some extreme inputs not accounted for it may happen
- in this study we have assumed that clipping does not occur and omitted it

1.4 Computing error for a linear layer

- we compute error $e^{(n)}$ for layer n
- in the quantized model a linear layer consists of the linear multiplication and the rescaling
- the error is the difference between base and the quantized model reverted back to floats
- at layer n: $\Delta x^{(n+1)} = |x^{(n+1)} - \hat{x}^{(n+1)}|$
  - base model: $x^{(n)} \xrightarrow{\text{linear}} z^{(n)} \xrightarrow{=} x^{(n+1)}$
  - quantized model: $x_{int}^{(n)} \xrightarrow{\text{linear}} z_{int}^{(n)} \xrightarrow{\text{rescaling}} x_{int}^{(n+1)}$
  - quantized model reversed to floats: $\hat{x}^{(n)} \xrightarrow{\text{linear}} \hat{z}^{(n)} \xrightarrow{\text{rescaling}} \hat{x}^{(n+1)}$
- we name the error bound: $e^{(n)}$

Result

$$\boxed{\Delta x^{(n+1)} \leq e^{(n+1)} = \underbrace{\frac{S_x^{(n)}}{2} \cdot \left\|W^{(n)}\right\|_\infty}_{\text{accumulated error}} + \underbrace{\frac{S_W^{(n)}}{2} \cdot \left\|x^{(n)}\right\|_1}_{\text{weight quant}} + \underbrace{\frac{S_W^{(n)} \cdot S_x^{(n)}}{4} \cdot d_{in}^{(n)}}_{\text{cross term}} + \underbrace{\frac{S_x^{(n+1)}}{2}}_{\text{rescaling}}} \tag{4}$$

with $d_{in}^{(n)}$ the input dimension of layer n

Proof In annex.

Remark

- Bound on error $\Delta x^{(n+1)}$ depends on previous input $x^{(n)}$
- Some implementations have global quantization $S_x^{(n)} = S_W^{(n)} = S$
- In this case, we can simplify our bounds to

$$\Delta x^{(n+1)} \leq \frac{S}{2} \left(\left\|x^{(n)}\right\|_1 + \left\|W^{(n)}\right\|_\infty + 1 \right) \frac{S^2}{4} \cdot d_{in}^{(n)}$$

  with $d_{in}^{(n)}$ the input dimension of layer n

1.5 Total error propagation

- The error from previous layer is amplified by matrix multiplication

The total error at layer L satisfies:

$$\boxed{\left\|\Delta x^{(L)}\right\|_\infty \leq \sum_{k=1}^{L} \left(\prod_{\ell=k+1}^{L} \left\|W^{(\ell)}\right\|_\infty \right) \cdot e^{(k)}} \tag{5}$$

Proof By induction. In annex. (TODO)

2 Augmenting models with non-linear layers

$$x^{(0)} \xrightarrow{\text{quantization}} \hat{x}^{(0)} \xrightarrow{\text{linear}} \hat{z}^{(0)} \xrightarrow{\text{rescaling}} \hat{x}^{(1)} \rightarrow \hat{z}^{(1)} \rightarrow \ldots \rightarrow \hat{z}^{(L)}$$

2.1 ReLU

ReLU does not bring new approximation since it works fine on integers.

- Case 1: Both neurons are active -> Effective error pass through unchanged
- Case 2: Both neurons are inactive -> Effective error is killed and is 0
- Case 3: Activation flip (one active and one inactive) -> Effective error is reduced

2.2 Others

- GELU, SiLU, Softmax, LayerNorm
- not computable on integers and need to be approximated
- this adds a new source of errors

- recent zkml works uses lookup tables for speed and precision improvement
- see NanoZK who boasts really low error on their lookups

3 Annexes

3.1 Proof of error bounds on a single layer

$$\boxed{\Delta x^{(n+1)} \leq \underbrace{W^{(n)} \cdot \frac{S_x^{(n)}}{2}}_{\text{accumulated error}} + \underbrace{\frac{S_W^{(n)}}{2} \cdot x^{(n)}}_{\text{weight quant}} + \underbrace{\frac{S_W^{(n)} \cdot S_x^{(n)}}{4}}_{\text{cross term}} + \underbrace{\frac{S_x^{(n+1)}}{2}}_{\text{rescaling}}}$$

3.1.1 Rescaling error

Result

$$\Delta x^{(n+1)} = |\Delta z^{(n)} - \Delta S_x^{(n+1)}|$$

Proof

$$\begin{aligned}
\Delta x^{(n+1)} &= \left\| x^{(n+1)} - \hat{x}^{(n+1)} \right\|_\infty \\
&= \left\| z^{(n)} - x_{int}^{(n+1)} \cdot S_x^{(n+1)} \right\|_\infty \\
&= \left\| z^{(n)} - round\left(\frac{\hat{z}^{(n)}}{S_x^{(n+1)}} \right) \cdot S_x^{(n+1)} \right\|_\infty \\
&= \left\| z^{(n)} - round\left(\frac{\hat{z}^{(n)}}{S_x^{(n+1)}} \right) \cdot S_x^{(n+1)} \right\|_\infty
\end{aligned}$$

Error due to rounding is represented by $\Delta S_x^{(n+1)}$

$$\begin{aligned}
&= \left\| z^{(n)} - \left(\frac{\hat{z}^{(n)}}{S_x^{(n+1)}} \cdot S_x^{(n+1)} + \Delta S_x^{(n+1)} \right) \right\|_\infty \\
&= \left\| z^{(n)} - \hat{z}^{(n)} - \Delta S_x^{(n+1)} \right\|_\infty \\
&= \left\| \Delta z^{(n)} - \Delta S_x^{(n+1)} \right\|_\infty \\
&\leq \left\| \Delta z^{(n)} \right\|_\infty + \left\| \Delta S_x^{(n+1)} \right\|_\infty \\
&\leq \left\| \Delta z^{(n)} \right\|_\infty + \frac{S_x^{(n+1)}}{2}
\end{aligned}$$

3.1.2 Accumulated error $\Delta z^{(n)}$

- we define $\hat{W}^{(n)} = W^{(n)} + \Delta W^{(n)}$
- we define $\hat{x}^{(n)} = x^{(n)} + \Delta x^{(n)}$

$$
\begin{aligned}
\Delta z^{(n)} &= \left\| z^{(n)} - \hat{z}^{(n)} \right\|_\infty \\
&= \left\| W^{(n)} \cdot x^{(n)} - \hat{W}^{(n)} \cdot \hat{x}^{(n)} \right\|_\infty \\
&= \left\| W^{(n)} \cdot x^{(n)} - (W^{(n)} + \Delta W^{(n)}) \cdot (x^{(n)} + \Delta x^{(n)}) \right\|_\infty \\
&= \left\| \Delta W^{(n)} \cdot x^{(n)} + W^{(n)} \cdot \Delta x^{(n)} + \Delta W^{(n)} \cdot \Delta x^{(n)} \right\|_\infty \\
&\leq \left\| \Delta W^{(n)} \cdot x^{(n)} \right\|_\infty + \left\| W^{(n)} \cdot \Delta x^{(n)} \right\|_\infty + \left\| \Delta W^{(n)} \cdot \Delta x^{(n)} \right\|_\infty \\
&\leq \max_i(|(\Delta W^{(n)} \cdot x^{(n)})_i|) + \max_i(|(W^{(n)} \cdot \Delta x^{(n)})_i|) + \max_i(|(\Delta W^{(n)} \cdot \Delta x^{(n)})_i|) \\
&\leq \max_i(|\sum_j \Delta W_{ij}^{(n)} \cdot x_j^{(n)})|) + \max_i(|\sum_j W_{ij}^{(n)} \cdot \Delta x_i^{(n)}|) + \max_i(|\sum_j \Delta W_{ij}^{(n)} \cdot \Delta x_i^{(n)}|) \\
&\leq \max_i(\sum_j |\Delta W_{ij}^{(n)}| \cdot |x_j^{(n)}|) + \max_i(\sum_j |W_{ij}^{(n)}| \cdot |\Delta x_i^{(n)}|) + \max_i(\sum_j |\Delta W_{ij}^{(n)}| \cdot |\Delta x_i^{(n)}|) \\
&\leq \max_i(\sum_j |\Delta W_{ij}^{(n)}| \cdot |x_j^{(n)}|) + \max_i(\sum_j |W_{ij}^{(n)}| \cdot |\Delta x_i^{(n)}|) + \max_i(\sum_j |\Delta W_{ij}^{(n)}| \cdot |\Delta x_i^{(n)}|) \\
&\leq \frac{S_W^{(n)}}{2} \max_i(\sum_j |x_j^{(n)}|) + \frac{S_x^{(n)}}{2} \max_i(\sum_j |W_{ij}^{(n)}|) + \sum_j \frac{S_x^{(n)}}{2} \frac{S_W^{(n)}}{2} \\
&\leq \frac{S_W^{(n)}}{2} \cdot \left\| x^{(n)} \right\|_1 + \frac{S_x^{(n)}}{2} \cdot \left\| W^{(n)} \right\|_\infty + \frac{S_W^{(n)} \cdot S_x^{(n)}}{4} \cdot d_{in}^{(n)}
\end{aligned}
$$

with $d_{in}^{(n)}$ the input dimension of layer n

3.1.3 Final result

$$
\Delta x^{(n+1)} \leq \frac{S_W^{(n)}}{2} \cdot \left\| x^{(n)} \right\|_1 + \frac{S_x^{(n)}}{2} \cdot \left\| W^{(n)} \right\|_\infty + \frac{S_W^{(n)} \cdot S_x^{(n)}}{4} \cdot d_{in}^{(n)} + \frac{S_x^{(n+1)}}{2}
$$

with $d_{in}^{(n)}$ the input dimension of layer n